COLIN NGUYEN

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ML researcher and AI engineer working on evaluation under uncertainty: how LLM agents and health-sensing models are measured, when a risk-control guarantee holds, and how to train models efficiently. Background in statistics.

INDUSTRY EXPERIENCE

Athena Money — Founding AI Engineer *Nov 2025 – Present*

- Built the team's multi-turn LLM evaluation platform for regulated mortgage underwriting (scenario graders, LLM-as-judge rubrics, per-turn scoring, an LLM user simulator); cut the supervisor evaluation run from about 55 minutes to 15.
- Took run-to-run noise out of agent evaluation: field-level consensus across trials, confidence calibration and regression gates in tiered CI on Temporal; an audit found eight framework and grader bugs, fixed in five days.
- Built an experiments platform comparing extraction and judgement approaches on regulated documents (LLM-as-judge, field-level agreement across runs, QC datasets); it later fed the evaluation framework.

Zeta Global — Data Science & GenAI Intern *Jun 2025 – Aug 2025*

- Cut manual workflow setup by 70% with a LangGraph supervisor agent that routes tasks to specialised sub-agents.

NationBuilder — AI Research Intern *Jan 2025 – Nov 2025*

- Ranked document relevance at 93% classification accuracy with a RAG pipeline (all-MiniLM-L6-v2 embeddings, FAISS), so irrelevant PDFs are skipped before processing.

RESEARCH EXPERIENCE

Evaluation of Mobile-Health Receptivity Models — with R. Benzo (OSU) and P. Washington (UCSF) *Jun 2026 – Present*

- Showed that pooled AUC, the standard metric for models predicting when someone will answer a health prompt, is dominated by between-person comparisons across three cohorts, so it mostly rewards knowing who tends to respond.
- Pre-registered a personalisation study (OSF) and trained a self-supervised GRU encoder on minute-level wearable data to test what its embeddings carry; results in the manuscript in preparation.

Risk-Controlled Decision Gates under Option-Count Shift — with Prof. Eric Suess, CSU East Bay *Oct 2026 – Present*

- Showed a Learn-then-Test abstention guarantee (5% risk, 90% confidence) fails under option-count shift, 5 to 150 options: 7.2% realised risk on a public decision model and, by the same mechanism, on mine; pre-registered.
- Fine-tuned SmolLM2-135M/360M into decision models (anchored listwise loss, token-trie readout) matching the hosted model at every option count; per-count calibration proved underpowered, rank-normalisation leaky.

Efficient Training Methods for Language Models — with Prof. Eric Suess, CSU East Bay *Sep 2026 – Oct 2026*

- Raised SmolLM2's trainable step sixteenfold (8,192 to 131,072 tokens) in 71% of baseline memory with five stacked techniques; the 17–22% throughput cost is all checkpointing.
- Hand-wrote a block-wise 8-bit AdamW that matched torchao's optimizer state to the byte, and showed it equivalent to fp32 across six seeds at two model sizes under a pre-registered TOST margin.

The Ohio State University — Lifestyles Lab (PI: R. Benzo) *Jun 2026 – Present*

- Co-leading the analysis plan validating consumer-wearable sleep measurement against research-grade actigraphy in cancer survivors (night-matched Bland–Altman, mixed models, TOST); migrated the lab's data tool to the Google Health API.

Systematic Review: Computer Vision for Home-Video Phenotyping of ASD and ADHD — UCSF *Oct 2025 – Present*

- Screened studies for a ~55-study review (lead: Dr. Abdur Rasool; PI: Dr. Peter Washington); responsible for the meta-regression of model performance on video length, setting, validation and ground truth. *Manuscript in preparation.*

SELECTED PROJECTS

NBA Catch Decisions — independent, on public player-tracking data *Oct 2026*

- Built a catch-level dataset from the only public NBA tracking data (621 games, 2015–16) and showed players trade a foot of space at the catch for about eight seconds of shot clock; every test pre-registered, failures kept.

SKILLS

Research methods: evaluation methodology, pre-registration, equivalence testing (TOST, Bland–Altman), conformal prediction and risk control (Learn-then-Test), calibration, survival analysis (Kaplan–Meier, Cox), Bayesian shrinkage, clustered bootstrap
ML / DL: PyTorch, fine-tuning and PEFT (LoRA, QLoRA), quantised training (torchao), vLLM, scikit-learn, TensorFlow
LLM systems: LangGraph, Google ADK, Temporal, RAG, LLM-as-judge evaluation, coding-agent workflows (Claude Code), vector search (FAISS)
Languages / Tools: Python, R, SQL, Kotlin, Swift, Java, Go; AWS, Google Cloud, Docker, GitHub Actions, Spark, Databricks

EDUCATION AND PATENTS

University of California, Santa Barbara *Sep 2022 – Jun 2025*
B.S. in Statistics & Data Science, Minor in Mathematics *Santa Barbara, CA*

- "Adaptive Neural Content Generation" (Zeta Global), U.S. Provisional App. No. 63/943,780, filed Dec 2025; first-named inventor.